

# Jyot Antani, PhD

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**Expertise:** Microscopy, Single Particle Tracking, Microbial Biophysics,  
Phage-Bacteria Interactions, Phage Therapy, Phage Technology

## Experience & Education

|                       |                                                                                                                                                                                                                                                                                                                                                                                                        |
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| Jun 2021-<br>Now      | Postdoctoral Research Associate, Yale University (US)<br><a href="#">Center for Phage Biology &amp; Therapy</a> , Yale University<br><a href="#">Quantitative Biology Institute</a> , Yale University<br><a href="#">Paul Turner Lab</a> , Department of Ecology and Evolutionary Biology<br>In collaboration with <a href="#">Thierry Emonet Lab</a> , Molecular, Cellular, and Developmental Biology |
| Aug 2016-<br>May 2021 | PhD, <a href="#">Pushkar Lele Lab</a> , Department of Chemical Engineering, Texas A&M University (US)<br>Dissertation title: Sensory Functions of the Bacterial Flagellar Motor                                                                                                                                                                                                                        |
| Jul 2012-<br>May 2016 | Bachelor of Technology with Honours in Chemical Engineering<br>Indian Institute of Technology Bombay (IIT Bombay), Mumbai (India)                                                                                                                                                                                                                                                                      |

**Advanced Microscopy Training:** [Optical Microscopy & Imaging in the Biomedical Sciences \(OMIBS\)](#)  
Marine Biological Laboratory, Woods Hole Massachusetts (US)

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|----------|-------------------------------------------|
| Aug 2023 | Research Facilitator (Teaching Assistant) |
| Aug 2022 | Course Participant                        |

- Basic principles of microscopy from academic and industrial microscope builders and inventors including...

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|-------------------------------------------------------|--------------------------------------------------------------|
| Light path, image formation, and detection            | Photophysics of fluorescence, dyes, and fluorescent proteins |
| Dynamic Imaging Techniques (FRET, FLIM, FCS)          | Point Spread Functions, Deconvolution, & Confocal Microscopy |
| Multiphoton & Light Sheet Microscopy; Adaptive Optics | Super-resolution Techniques: STED, STORM, SIM                |
- Hands-on practice with over 40 microscopes including the following modalities...

|                           |                                          |                                    |
|---------------------------|------------------------------------------|------------------------------------|
| Phase-contrast Microscopy | Differential Interference Contrast (DIC) | Polarized Light Microscopy         |
| Epifluorescence           | Forster Resonance Energy Transfer        | Fluorescence Lifetime Imaging      |
| Confocal Microscopy       | Total Internal Reflection Fluorescence   | Deconvolution                      |
| Multiphoton Microscopy    | Light Sheet Microscopy                   | Super-resolution: STED, STORM, SIM |

## Publications

[Google Scholar Link](#)

\* indicates equal contribution

Oromí-Bosch A\*, [Antani JD\\*](#), Turner PE (2023) Developing Phage Therapy That Overcomes the Evolution of Bacterial Resistance. *Annu Rev Virology*. doi: [10.1146/annurev-virology-012423-110530](https://doi.org/10.1146/annurev-virology-012423-110530)

[Antani JD](#), Gupta R, Lee AH, Rhee KY, Manson MD, Lele PP (2021) Mechanosensitive recruitment of stator units promotes binding of the response regulator CheY-P to the flagellar motor. *Nature Communications*. doi: [10.1038/s41467-021-25774-2](https://doi.org/10.1038/s41467-021-25774-2)

[Antani JD](#), Sumali AX, Lele TP, Lele PP (2021) Asymmetric random walks reveal that the chemotaxis network modulates flagellar rotational bias in *Helicobacter pylori*. *eLife*. doi:[10.7554/eLife.63936](https://doi.org/10.7554/eLife.63936)  
*This work helped secure an R01 research grant totaling over \$1.3 million from the National Institute of General Medical Sciences, National Institute of Health.* <https://www.eurekalert.org/news-releases/945164>

Katiyar A\*, [Antani JD\\*](#), McKee BP, Gupta R, Lele PP, Lele TP (2021) A method for direct imaging of x-z cross-sections of fluorescent samples. *Journal of Microscopy*. doi:[10.1111/jmi.12965](https://doi.org/10.1111/jmi.12965)

Wong GCL, [Antani JD](#), et al. (2021) Roadmap on emerging concepts in the physical biology of bacterial biofilms: from surface sensing to community formation. *Physical Biology*. doi:[10.1088/1478-3975/abdc0e](https://doi.org/10.1088/1478-3975/abdc0e)

Antani JD, Shaji A, Gupta R, Lele PP. Reassessing the Standard Chemotaxis Framework for Understanding Biased Migration in *Helicobacter pylori*. *Annu Rev Chem Biomol Eng*. doi:10.1146/annurev-chembioeng-100722-114625

Ford KM, Antani JD, Nagarajan A, Johnson MM, Lele PP (2018) Switching and torque generation in swarming cells of *E. coli*. *Frontiers in Microbiology*, 9, 2197. doi:10.3389/fmicb.2018.02197

Katiyar A, Zhang J, Antani JD, Yu Y, Lele PP, Reinhart-King CA, Sniadecki NJ, Roux KJ, Dickinson RB, Lele TP (2022) The Nucleus Bypasses Obstacles by Deforming Like a Drop with Surface Tension Mediated by Lamin A/C. *Advanced Science*. doi: 10.1002/advs.202201248

## Publications in preparation

Antani JD, Ward T, Emonet T, Turner PE. A general method to quantify adsorption of single viruses to bacterial cells.

Antani JD, Theroux A, Emonet T, Turner PE. Evolutionary consequences of evolution of motile bacteria against a flagellum-binding virus. Media mention of this work in the *ASM Microcosm magazine* (link)

## Peer Reviews (Independent)

[Publons / WebOfScience Link](#)

|                                        |                             |                                                     |
|----------------------------------------|-----------------------------|-----------------------------------------------------|
| <i>Microbiology Spectrum</i> (18)      | <i>Physical Biology</i> (4) | <i>Applied &amp; Environmental Microbiology</i> (1) |
| <i>Archives of Microbiology</i> (11)   | <i>PeerJ</i> (10)           | <i>STAR Protocols</i> (2)                           |
| <i>Frontiers in Microbiology</i> (2)   | <i>Biosystems</i> (2)       | <i>Frontiers in Toxicology</i> (1)                  |
| <i>European Biophysics Journal</i> (1) | <i>PLoS One</i> (2)         | <i>APL Bioengineering</i> (3)                       |
| <i>Cancer Reports</i> (1)              |                             |                                                     |

## Technical Skills

|                                         |                           |                                        |
|-----------------------------------------|---------------------------|----------------------------------------|
| Perfusion & Thermal Control Experiments | Prototyping & 3D-printing | Optical Microscopy: advanced expertise |
| Bacteria & Phage Culturing, Cloning     | Microfluidics Experiments | Image Analysis Automation              |
| Genome Analysis/Bioinformatics          | Assay Development         | Single Particle Tracking               |

## Computer Proficiency / Computational Skills

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|------------------------------|--------------------------------------------------------------------|
| Image Processing & Analysis: | MATLAB, Python, ImageJ – {Highlight: particle-tracking algorithms} |
| Genetic Analysis:            | Molecular Biology {SnapGene, Benchling}, Bioinformatics            |
| Design & Development:        | SolidWorks, AutoCAD, LabVIEW,                                      |
| Simulation Packages:         | ANSYS Fluent, Aspen Plus, OpenFOAM, COMSOL, CHARMM, VMD            |
| Programming:                 | MATLAB, Python, C, C++, Fortran, HTML                              |

## Special mention: Yale Biotech Club Datathon 2022

- Performed analysis on single-cell RNAseq data from patients of Colorectal Cancer and Inflammatory Bowel Disease
- Used [scanpy](#) to study the variability in expression of various genes and identify marker genes associated with the two diseases

## Honors & Awards

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|------|--------------------------------------------------------------------------------------------------------------|
| 2023 | Top Reviewer 2023, <i>Microbiology Spectrum</i> – an American Society for Microbiology (ASM) Journal         |
| 2023 | Best Poster Presentation Award, <i>Yale Postdoctoral Association Annual Symposium 2023</i> , Yale University |
| 2022 | <i>BioImaging North America (BINA)</i> Professional Development Award - <i>Chan Zuckerberg Initiative</i>    |
| 2022 | <i>Histochemical Society (HCS)</i> Travel Award                                                              |
| 2020 | CIRTL Associate Certificate, Center for the Integration of Research, Teaching, and Learning                  |
| 2015 | Best Poster Award, <i>AMAT</i> seminar, IIT Bombay                                                           |
| 2015 | Undergraduate Research Award URA01, IIT Bombay                                                               |
| 2012 | All India Rank 1427 in Joint Entrance Examination for engineering, among 479,651 participants                |
| 2008 | National Talent Search Scholarship, NCERT, India                                                             |

## Invited Talks/Seminars

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|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nov 2023 | Microbial Treasure Hunt: How Bacteria and Viruses Find their Haven. Departmental Seminar, Chemical Engineering, Indian Institute of Technology Bombay (India) |
| Mar 2023 | Physics of phage adsorption to bacterial surface. HHMI EPI Meeting, University of California San Diego (US)                                                   |
| Oct 2021 | Force-regulation of CheY-P-binding to the bacterial flagellar motor. Howard Berg Lab, Harvard University (US)                                                 |

## Conference Presentations

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Antani JD, Ward TJ, Emonet T, Turner PE. Quantitative biology of bacterial virus attachment to host surfaces.

May 2023 | Yale Postdoctoral Association Symposium, Yale University, New Haven (US)  
Sep 2022 | Annual Microbiology Retreat, Yale University, New Haven (US)  
Jun 2022 | *Microbe*- Annual meeting of the American Society for Microbiology, Washington, D.C. (US)  
May 2022 | Yale Postdoctoral Association Symposium, Yale University, New Haven (US)

Antani JD, Theroux A, Emonet T, Turner PE. Bacterial Evolution Of Resistance To A Flagellotropic Phage.

Sep 2023 | Annual Microbiology Retreat, Yale University, New Haven (US)  
Jun 2023 | *Microbe*- Annual meeting of the American Society for Microbiology, Houston (US)

Antani JD, Lele PP. Modulation of chemotactic signaling in bacteria by mechanical forces.

Mar 2019 | ChEGSA Annual Symposium, Texas A&M University, College Station (US)  
Oct 2018 | American Institute of Chemical Engineers (AIChE) Annual Meeting, Pittsburgh (US)  
Dec 2017 | Texas Branch of American Society for Microbiology, 2017 Fall Meeting, College Station (US).

Antani JD, Lele PP. Asymmetric Swimming and Chemotaxis in *Helicobacter pylori*.

Jan 2021 | Bacterial Locomotion And Signal Transduction (BLAST) XVI Meeting, Zoom (online)  
Feb 2020 | Biophysical Society (BPS) Annual Meeting, San Diego (US).  
Jan 2019 | Bacterial Locomotion And Signal Transduction (BLAST) XV Meeting, New Orleans (US)

Antani JD, Venkatesh KV. Analysis of Glucose Homeostasis in Humans.

Nov 2015 | Futuristic Approach for Alternatives to Animal Testing [CEFIPRA](#) seminar, Mumbai (India)

## Current and Doctoral Research

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| May 2021-<br>Now      | <p>Postdoctoral Research: <b>Host-virus Interaction Biophysics in Microbes</b><br/><i>Evolution of target search by bacteriophages on host bacterial surfaces</i></p> <ul style="list-style-type: none"><li>Established a protocol for labeling phages (the viruses of bacteria) with a lysine-binding fluorescent dye</li><li>Visualizing labeled phage (bacterial virus) particles attaching to host cells, through epifluorescence and HILO microscopy</li><li>Set up MATLAB-based statistical analysis of single phage trajectories to quantify attachment dynamics</li><li>Employing Bayesian inference to estimate the diffusive states of phage particles from single trajectories</li><li>Developing a quick high-throughput assay to quantify phage adsorption, applicable for emergency phage therapy</li><li>Comparing the attachment dynamics of ancestral and coevolved phage-host pairs</li></ul> <p><i>Evolution of resistance in E. coli against a flagellotropic phage</i></p> <ul style="list-style-type: none"><li>Evolved mutants resistant against phage <math>\chi</math>, a virus that uses rotating bacterial flagella as receptors</li><li>Trained an undergraduate to carry out the experimental evolution protocol, mentored them in writing a successful fellowship application about this project</li><li>Characterizing the genotype and phenotype of the evolved mutants to determine the evolutionary consequences</li></ul> <p><i>Grant-Writing</i></p> <ul style="list-style-type: none"><li>Wrote several fellowship and grant applications to fund the original research projects</li></ul> |
| Aug 2016-<br>May 2021 | <p>Doctoral Thesis: <b>Sensory Functions of the Bacterial Flagellar Motor</b><br/><i>Molecular mechanisms of host substrate sensing by bacterial flagellar motors</i></p> <ul style="list-style-type: none"><li>Determined the effect of mechanical forces on chemotactic output of the bacterial flagellar motor, with phase microscopy, optical tweezers, and TIRF imaging</li><li>Developed a biophysical model for the regulation of allosteric interactions by mechanical force</li></ul> <p><i>Motility, chemotaxis, and random spread of Helicobacter pylori, the carcinogenic pathogen</i></p> <ul style="list-style-type: none"><li>Devised a microscopy-based technique to experimentally determine the chemotactic activity of <i>H. pylori</i></li><li>Studied the biophysical principles governing chemotactic migration by this bacterial species which employs run-reversal motility – different from the traditionally studied <i>E. coli</i></li></ul> <p><i>Design and development of experimental and analytical framework</i></p> <ul style="list-style-type: none"><li>Designed 3D-printed imaging chambers for requirements in microscopy experiments, e.g., perfusion, imaging side-views</li><li>Wrote MATLAB-based image analysis codes for tracking motile bacteria, mammalian cell-nuclei, and fluorophores</li></ul>                                                                                                                                                                                                                                                                                |

## Additional Research Experience

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Jul 2015-  
Feb 2016 | Machine Learning Classification of Glucose Homeostasis in Diabetic and Healthy Individuals  
*Venkatesh Lab, Chemical Engineering, IIT Bombay*

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|                     | <ul style="list-style-type: none"> <li>Formulated a Hill function-based empirical biochemical model of body glucose concentration following a fixed food-intake, obtained least-squares fits to data acquired from 150+ individuals</li> <li>Constructed a classification tree based on fit-parameters using machine learning in MATLAB; established a statistical grouping among Diabetic and Healthy individuals</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| May-Jul 2015        | <p>A Novel Protocol for Coating Lipid Bilayers onto Silver Nanocubes<br/><i>Hung-Jen Wu Lab, Texas A&amp;M University</i></p> <ul style="list-style-type: none"> <li>Established a tip-sonication protocol to make small unilamellar vesicles of lipids</li> <li>Coated the unilamellar vesicles as supported lipid bilayers onto Silver nanocubes</li> <li>Employed a Localized Surface Plasmon Resonance-based nanocube sensor to monitor protein binding on lipid bilayers</li> <li>Analyzed binding kinetics of GM1 ganglioside with Cholera Toxin Subunit B</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| May 2014- Apr 2015  | <p>Simulation-Design of a Chemical Vapor Deposition Reactor<br/><i>Mahesh Tirumkudulu Lab, IIT Bombay</i></p> <ul style="list-style-type: none"> <li>Simulated fluid flow around a disk, successfully obtained the pressure coefficients observed experimentally in literature.</li> <li>Proposed an unconventional design for uniform coating of parylene polymer onto silicon micro-wafers in a micro-scale Chemical Vapor Deposition reactor.</li> <li>Modeled Parylene flow using ANSYS Fluent, enhanced the proposed design.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Jun 2013            | <p>Process Improvement and Technology in Jaggery-Making<br/><i>Sanjay Mahajani Lab, IIT Bombay</i></p> <ul style="list-style-type: none"> <li>Examined units under development for continuous production as well as recuperation in manufacturing Jaggery, the traditional non-centrifugal cane sugar</li> <li>Analyzed Jaggery samples for acidity, moisture content, water activity, density, sugar content and structure (XRD, ESEM)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Key Course Projects | <p>Determining Diffusion Coefficient of Microparticles using Particle-Tracking Algorithms<br/><i>Course: Colloidal Applications and Biomolecular Interactions in Life Science (Spring 2018)</i></p> <ul style="list-style-type: none"> <li>Employed particle-tracking algorithms in MATLAB to track the Brownian diffusion of submicron-sized particles recorded using phase-contrast microscopy</li> <li>Determined the diffusion coefficient of particles by calculating the average mean-squared displacement of an ensemble</li> </ul> <p>Proposal: Evolution of Bacterial Flagellum<br/><i>Course: Evolutionary Bioinformatics (Spring 2017)</i></p> <ul style="list-style-type: none"> <li>Proposed a study involving the comparison of flagellar genome across multiple bacterial species which has potential to aid in obstruction of pathogenic motility</li> <li>Carried out BLAST alignment analysis for nucleotide and protein sequences of flagellin (flagellar filament) gene as preliminary results</li> </ul> <p>Review: How Deep a Cell Can Sense<br/><i>Course: Cell Mechanics (Spring 2016)</i></p> <ul style="list-style-type: none"> <li>Carried out a literature survey and reported theoretical, in-vitro, and in-silico studies on quantified investigations about the range of mechanical signaling</li> </ul> <p>On-line Estimation of Concentrations in a Bio-Reactor<br/><i>Course: State Estimation (Fall 2015)</i></p> <ul style="list-style-type: none"> <li>Generated true states from a model based on substrate uptake, phosphorus repression and effect of seed culture on glyco-peptide antibiotic production</li> <li>Implemented the Simple, Extended, and Unscented versions of the Kalman Filter to estimate the concentrations</li> </ul> |

## Teaching & Mentoring Experience

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| Fall 2019 | <p>A hands-on workshop for image processing and single particle tracking in MATLAB<br/><i>Pushkar Lele Lab, Texas A&amp;M University</i></p> |
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### Undergraduate students mentored

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| Sep 2022- Now      | <p>Timothy Ward, Undergraduate Researcher, Yale University<br/>Attachment of bacteriophages to bacterial cells: the classical adsorption assays<br/>Best Poster Award winner at <a href="#">Yale Undergraduate Research Symposium</a>, March, 2023</p>                                                                                                        |
| Oct 2021- Oct 2022 | <p><a href="#">Austin Theroux</a>, STARS II Science &amp; Quantitative Reasoning Education Scholar, Yale University<br/>Evolution of motile bacteria against a bacteriophage that attaches through the flagellum<br/>Presented a poster about this work at <i>Microbe</i> ASM meeting at Washington DC in June, 2022</p>                                      |
| Dec 2019- Jan 2021 | <p><a href="#">Anita Sumali</a>, Gathright Scholar, Biomedical Engineering, Texas A&amp;M University<br/>MATLAB-based quantitative analysis of the run-reversal motility of <i>H. pylori</i>; co-author on <a href="#">Antani et al., eLife, 2021</a><br/>Currently enrolled in EnMed (MEng/MD dual degree program) at Texas Medical Center, Houston (US)</p> |
| Apr 2017- Apr 2018 | <p>AggiE_Challenge – Undergraduate Mentoring, Texas A&amp;M University<br/>Graduate-mentor to a multidisciplinary team of eight undergraduate students employing 3D-printing technology to design scaled-up bacterial models, in order to study the hydrodynamic effects of cell morphology on motility</p>                                                   |

Teaching Assistant at Texas A&M Chemical Engineering

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|-------------|--------------------------|
| Spring 2019 | Heat Transfer Operations |
| Spring 2017 | Bioprocess Engineering   |
| Fall 2016   | Bioprocess Engineering   |

## Teaching, Mentoring, & Management Training

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| Sep 2021-<br>Dec 2021 | Auburn University's Preparing Future Faculty Workshop<br>Interactive workshops consisting panel discussions from faculty and university administrators                                                                                                                                                                                                                                                      |
| Sep 2021              | <a href="#">Fundamentals of Inclusive Teaching</a><br><a href="#">Poorvu Center: Graduate and Postdoctoral Teaching Development, Yale University</a><br>A 4-part workshop on creating a respectful, supportive, equitable, and transparent classroom environment                                                                                                                                            |
| Aug 2018-<br>Dec 2019 | <a href="#">Academy for Future Faculty (AFF)</a> , Center for Teaching Excellence, Texas A&M University<br><i>Faculty Mentor: Dr. Micah Green, Texas A&amp;M University</i><br>A program designed for training graduate students as future faculty members, featuring weekly seminars and workshops, and anchored by faculty mentorship                                                                     |
| Jul-<br>Aug 2019      | An Introduction to Evidence-Based Undergraduate STEM Teaching<br><i>Muti-university Massive Open Online Course from the <a href="#">NSF-founded CIRTl network</a></i><br>Eight-week online course on <a href="#">edX.org</a> catering introduction to effective teaching strategies and the research that supports them. Attended local learning community meetings for the course at Texas A&M University. |
| Oct 2019              | Improving Research Mentoring Relationships – Introduction to Effective Mentorship<br><i>Center for the Improvement of Mentored Experiences in Research (CIMER)</i><br>1-day workshop at Texas A&M University, conducted by CIMER from University of Wisconsin-Madison                                                                                                                                       |

## Leadership & Management Experience

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| Jan 2022-<br>Now      | Communications Committee Manager and Coordinator, <a href="#">Yale Postdoctoral Association (YPA)</a><br>• Leading a team of volunteers to manage website, social media, and weekly + monthly newsletters for the YPA                                                                                                                                                                                                                                                                                                                                                                                       |
| Jun 2021              | <a href="#">People Management Skills</a> – Course, Chartered Institute of Personnel and Development, UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Oct 2018-<br>Aug 2020 | <a href="#">Secretary (2018-19) &amp; Chair (2019-20), Chemical Engineering Graduate Student Advisory Council</a><br><i>Artie McFerrin Department of Chemical Engineering, Texas A&amp;M University</i><br>• Designed the council by-laws as a founding member<br>• Set up channels of communication through surveys and anonymous feedback mechanisms, between grad students and administration in the department<br>• Addressed grad student welfare issues including salary, benefits, mental well-being, work-life balance, student-advisor relationships, facilities, career events, and social events |
| Apr 2014-<br>Mar 2015 | Competitions Manager, AZeotropy 2015<br><i>IIT Bombay's Intercollegiate Chemical Engineering Symposium</i><br>• Pioneered a new model-making competition <i>ChemETimer</i> based on time-precision<br>• Designed a presentation-based competition <i>Swaasthya</i> on engineering-applications to healthcare<br>• Designed <i>Cipher</i> , the online cryptic hunt which received a participation of 400+<br>• Actualized unconventional chemical engineering quizzing for <i>Chem-O-Philia</i> , national level quiz                                                                                       |

## Science Outreach Volunteering Experience

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| Mar 2023  | <a href="#">Tiny World for Little Humans: Teaching Quantitative Microscopy to Underserved Youth</a><br>• Taught quantitative microscopy through the use of paper microscopes ( <a href="#">foldscope.com</a> ) to middle school students at Fair Haven School, New Haven, Connecticut<br>• Outreach project funded by the <a href="#">MBL ROCS- Marine Biological Laboratory Regional Outreach and Communication in STEM</a> |
| Feb 2022- | <a href="#">Associate Writer &amp; Editor, LabLeaks</a>                                                                                                                                                                                                                                                                                                                                                                      |

- Sep 2022 | A biweekly mailing list summarizing recent noteworthy scientific articles to the general audience.
- Feb 2022 | [Exploring Science for Middle School Students](#)  
Presented doctoral research topic to middle school children in the New Haven County, Connecticut
- Jul 2020 | [Coronasurveys: Monitoring the Incidence of COVID-19 via Open Surveys](#)  
Volunteered services as the primary translator of the survey to the Gujarati language.